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Reflection Journal – Image Processing Fundamentals

This lab gave me a much better understanding of how computers represent and process images. Before completing the lab, I knew that images were made up of pixels, but I had never really thought about how a computer sees an image. What surprised me the most was learning that an image is simply a matrix of numerical values. Instead of seeing objects, people, or scenery, a computer only sees arrays of numbers that represent pixel intensities. It was especially interesting to separate the RGB channels and see how a color image is stored as three different layers of information. I also found it surprising that converting an image to grayscale significantly reduces the amount of data while still preserving most of the important visual information.

The mathematical operations performed during the lab helped me understand how visual effects are created. Adjusting brightness and contrast showed how simple changes to pixel values can dramatically change the appearance of an image. The neighborhood operations were even more interesting because they considered surrounding pixels rather than processing each pixel independently. By applying blur, sharpening, and edge detection filters, I was able to see how convolution kernels transform images and highlight specific features. The technique that was hardest for me to grasp was histogram equalization and CLAHE. Although I understood that both methods improve image contrast, it took me some time to understand how redistributing pixel intensities changes the histogram and enhances image details. After comparing the before-and-after results, I was able to better understand the purpose of these techniques.

Today's activities directly relate to the tools discussed in class, especially OpenCV and Pillow. During the lab, I used OpenCV to perform many of the image processing operations that we learned about in the lecture. The experience helped me understand that modern computer vision systems depend on many of these same foundational techniques. Image enhancement, filtering, edge detection, and geometric transformations are often used before machine learning models analyze images. The lab also helped me connect traditional image processing techniques to modern AI tools. While AI-powered applications may seem very advanced, they still rely on many of the same concepts explored in this lab.

These techniques have many real-world applications. In healthcare, image enhancement can improve medical scans and help doctors identify abnormalities more easily. In autonomous vehicles, image preprocessing can improve camera data before object detection systems analyze the environment. Manufacturing companies use image processing for quality control and defect detection, while smartphone cameras use similar techniques to improve photo quality.

automatically. Understanding these applications helped me see how image processing is used in many industries beyond computer science.

As an AI and Robotics student, I am interested in how traditional image processing can be combined with modern AI approaches. For example, a future robotics project could use OpenCV to improve image quality and reduce noise before a deep learning model performs object detection or classification. Combining traditional methods with machine learning could improve both accuracy and efficiency. I think understanding the fundamentals makes it easier to understand how more advanced computer vision systems work.

The topic that interests me most for further exploration is object detection and image segmentation. I would like to learn how computers move beyond basic image processing and begin identifying specific objects within an image. I am also interested in learning more about how vision-language models combine image understanding with natural language processing. One question I still have is how modern AI systems learn to recognize complex objects and scenes with such high accuracy compared to traditional methods. Overall, this lab strengthened my understanding of image processing and showed me how these foundational techniques connect to both traditional computer vision and modern AI applications.